

Lab Solution

Create a document `SOLUTION.md` with:

1. Cost Estimates:

- Link to Scenario 1 (baseline) estimate

<https://calculator.aws/#/estimate?id=4a7348b938778c703a2a7daaa49fb51eb865b58f>

- Link to Scenario 2 (optimized) estimate

<https://calculator.aws/#/estimate?id=5e296a174677973b12c3da07ec1c34ad8dad5f58>

- Comparison table showing cost differences

https://docs.google.com/spreadsheets/d/1iawI2FATjsO0Q7lG8q2MOFYZXHO-_vYG7suc8ezyOpQ/edit?gid=1313378920#gid=1313378920

2. Screenshots:

- Pricing Calculator estimate (both scenarios)

The screenshot displays the AWS Pricing Calculator interface. At the top, a green banner indicates "Successfully updated Amazon EC2 estimate." Below this, the page title is "My Estimate" with an "Edit" link. On the right, there are "Export" and "Share" buttons. The main content area is divided into two sections: "Estimate summary" and "Getting Started with AWS".

Estimate summary

Category	Cost
Upfront cost	0.00 USD
Monthly cost	87.62 USD
Total 12 months cost	1,051.44 USD
Includes upfront cost	

Getting Started with AWS

Buttons: Get started for free, Contact Sales

My Estimate

Buttons: Duplicate, Delete, Move to, Create group, Add support, Add service

Search: Find resources

Service Name	Status	Upfront cost	Monthly cost	Description	Region
Amazon EC2	-	0.00 USD	30.37 USD	-	US East (N. Vir...)
Amazon RDS for MySQL	-	0.00 USD	54.86 USD	-	US East (N. Vir...)
Amazon Simple Storage Service (S3)	-	0.00 USD	2.39 USD	-	US East (N. Vir...)

Ironhack Student Po... Quantity 0/0

aws pricing calculator Feedback Language: English Contact Sales Create an AWS Account

Estimate summary [Info](#)

Upfront cost	Monthly cost	Total 12 months cost
0.00 USD	62.26 USD	747.12 USD
		Includes upfront cost

Getting Started with AWS

[Get started for free](#)
[Contact Sales](#)

My Estimate

[Duplicate](#)
[Delete](#)
[Move to](#)
[Create group](#)
[Add support](#)
[Add service](#)

☐	Service Name	Status	Upfront cost	Monthly cost	Description	Region
☐	Amazon EC2	-	0.00 USD	4.99 USD	-	US East (N. Vir...
☐	Amazon RDS for MySQL	-	0.00 USD	54.86 USD	-	US East (N. Vir...
☐	Amazon Simple Storage Service (S3)	-	0.00 USD	2.41 USD	-	US East (N. Vir...

Acknowledgement
 AWS Pricing Calculator provides only an estimate of your AWS fees and doesn't include any taxes that might apply. Your actual fees depend on a variety of factors, including your actual usage of AWS services. [Learn more](#)

- Billing alarm configuration

AWS Notification - Subscription Confirmation 收件箱 x

AWS Notifications <no-reply@sns.amazonaws.com> 15:13 (0分钟前) ☆ 😊 ↩ ⋮

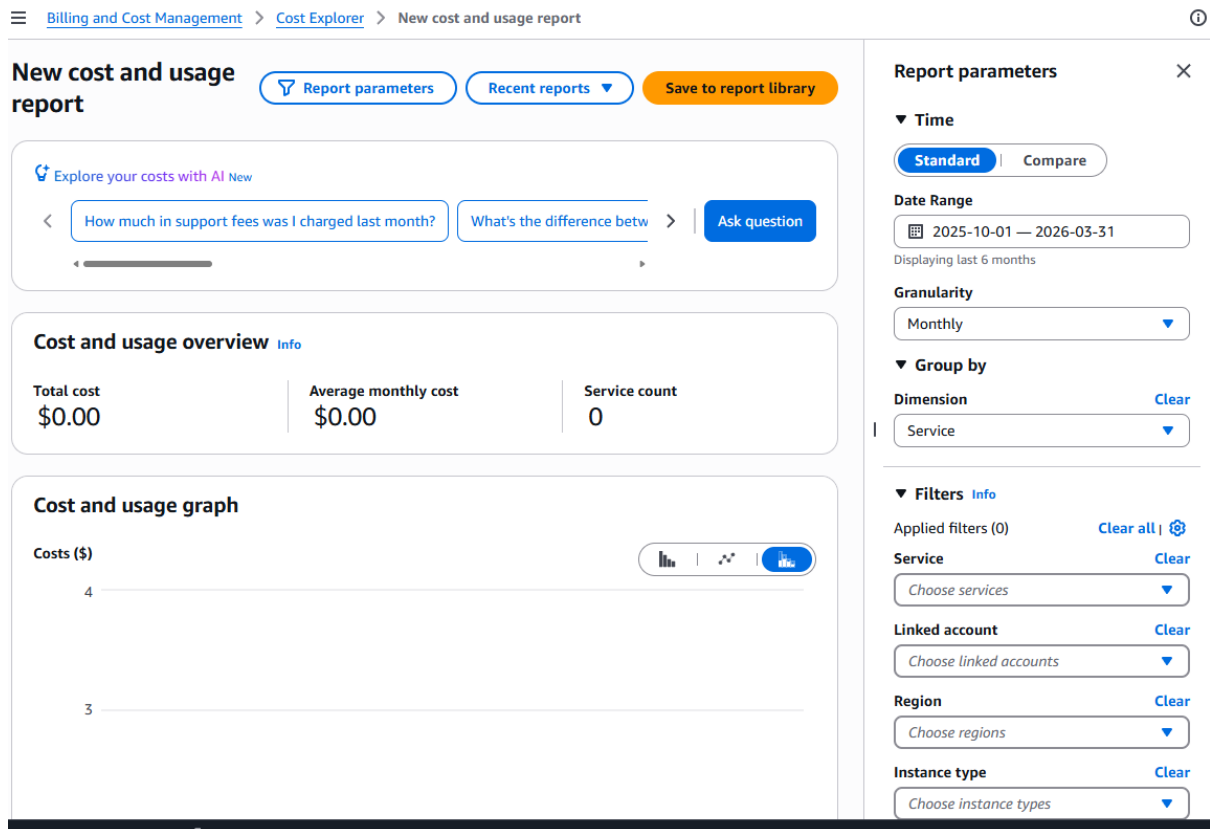
发送至 我 ▾

You have chosen to subscribe to the topic:
arn:aws:sns:us-east-1:047612973365:billing-alerts-0476-1297-3365

To confirm this subscription, click or visit the link below (If this was in error no action is necessary):
[Confirm subscription](#)

Please do not reply directly to this email. If you wish to remove yourself from receiving all future SNS subscription confirmation requests please send an email to [sns-opt-out](#)

- Cost Explorer views



****3. Analysis:****

Answer these questions:

- ****What service is the most expensive in Scenario 1?****

Answer: The most expensive service is Amazon RDS for MySQL at \$54.86 per month, accounting for approximately 62.6% of the total cost.

- ****How much money did you save in Scenario 2? What percentage savings?****

Answer: * **Total Savings:** $\$87.62 - \$62.26 = \$25.36$

- **Percentage Savings:** $(\$25.36 / \$87.62) \times 100\% \approx 28.9\%$

- ****What are the trade-offs of running the EC2 instance only during business hours?****

Answer: * **Availability:** The web application will be completely inaccessible to users during nights and weekends.

- **Maintenance:** Automated tasks, backups, or security patches that usually run during "off-peak" hours cannot be executed while the instance is stopped.
- **Global Reach:** This strategy only works for internal tools or localized apps where all users reside in the same time zone.

- ****If your website has unpredictable traffic, which pricing model would you use and why?****

Answer: I would use the **On-Demand** pricing model combined with **Auto Scaling**.

- **Why:** On-Demand allows you to pay only for the compute capacity you consume by the second. By using Auto Scaling, the system can automatically add instances during traffic spikes and terminate them when traffic drops, ensuring you don't pay for idle resources while maintaining site performance.

- ****How would you further optimize costs without impacting performance?****

Answer: 1. **Savings Plans or Reserved Instances (RI):** If the database must run 24/7, committing to a 1-year or 3-year term can reduce costs by up to 72% compared to On-Demand.

2. **Rightsizing:** Monitor the CloudWatch metrics; if the t3.medium CPU utilization is consistently low, I would downgrade to a t3.small or t3.micro.

3. **Storage Optimization:** Regularly delete old EBS snapshots and RDS backups that are no longer needed.

4. **S3 Lifecycle Policies:** Move older, infrequently accessed data to **S3 Glacier Deep Archive** to significantly reduce storage costs.
